Study of Noninvasive Glucose Monitoring Based on FBG for Ballistocardiogram

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***Abstract*—In this article, we propose and validate a noninvasive glucose monitoring method based on fiber optic sensing for heart rate variability ballistocardiography (HRVBCG). The aim was to monitor changes in blood glucose levels by analyzing the HRV parameters derived from the BCG signal. To avoid noise interference with BCG signal extraction, we introduced a variational mode decomposition parameter selection algorithm (S-VMD), which incorporates improvements in center frequency and peak index. Compared with electrocardiogram (ECG) signals, the maximum relative error in BCG signal feature point localization was 2.63%, and the maximum relative error between HRVECG and HRVBCG parameters was 7.5659%. These results indicate a high degree of consistency in the localization of characteristic points between BCG and ECG signals. This demonstrates that the proposed VMD parameter selection algorithm can effectively extract the BCG signal and retain the signal characteristic point data. To carry out the next step of correlation analysis of blood glucose value and HRV change amount. Experimental results revealed that when volunteers consumed food, their blood glucose levels increased, accompanied by an acceleration in heart rate and a decrease in HRV parameters such as standard deviation of NN (SDNN) intervals and root mean square of successive differences (RMSSDs) began to decrease. The overall correlations between the fit regression models for the HRV parameters and the amount of change in blood glucose values were all greater than 0.85, and the mean square errors were all less than 0.12. These findings suggest that the inversion of blood glucose parameters through BCG signal monitoring is feasible in the field of noninvasive blood glucose monitoring.**

***Index Terms*—Ballistocardiogram, feature point calibration, fiber Bragg grating, noninvasive glucose monitoring, variational modal decomposition.**

I. Introduction

BLOOD glucose monitoring is critical to health [1]. Currently used invasive blood measurement methods, although low-cost and reliable, suffer from invasiveness, pain, and infection [2]. To avoid these risks, researchers have turned to analyzing the correlation between human metabolic substances or vital signals and glucose. For example, Lin et al. [3], Park et al. [4], and Zhang et al. [5] monitored glucose levels through saliva, tears, and urine, while Kapur et al. [6] assessed blood glucose through volatile organic compound concentrations in breath samples with an accuracy of 86.6%. These noninvasive methods, although effective, still face problems of standardized collection and variability. Li et al. [7] and Nakazawa et al. [8] developed a noninvasive blood glucose monitoring device based on metabolic heat and infrared spectroscopy with an accuracy of more than 90%, but it is highly affected by the ambient temperature. Hammour and Mandic [9] used an in-ear device based on PPG for blood glucose measurements, with an accuracy of up to 0.82. Kajisa et al. [10] and Coronado-Reyes et al. [11] monitored blood glucose changes by electrocardiogram (ECG) with an accuracy of more than 90%, but the need to apply couplers and the large size of the device made it unsuitable for home use. Based on this, researchers started to introduce human BCG signals to calculate HRV parameters for glucose monitoring.

In practice, BCG signals are usually acquired by electronic or optical sensors. Electrical sensors have problems such as a complex structure and electromagnetic interference, and are less comfortable to wear. Common optical sensors include PPG [12], LIDAR [13], and fiber-grating [14], of which PPG and LIDAR require specific lighting conditions and are more costly. In contrast, fiber-grating sensors are smaller and more stable and are more suitable for daily noninvasive glucose monitoring. With the development of BCG detection technology, the corresponding signal processing algorithms need to be gradually optimized. Jiang et al. [15] used empirical modal decomposition with independent component analysis to remove high-frequency noise, but there are problems of mode aliasing and endpoint effects. Wen et al. [16] used wavelet transform and root-mean-square filtering to remove the noise, but it is susceptible to the number of decomposition layers. In contrast, VMD shows better performance in nonstationary signal analysis. However, in order to fully utilize the advantages of this algorithm, the proper number of modes K and penalty parameter α need to be predetermined.

Aiming at the existing noninvasive blood glucose monitoring technology, there are problems such as low precision, long use time, cumbersome operation, and so on. In this article, a noninvasive blood glucose monitoring method based on the BCG signal is proposed. By embedding the fiber optic sensor into the strap, the strain signal generated by the vibration of the chest wall is captured, and the signal is processed using a fiber optic demodulator. To remove the noise, an improved S-VMD algorithm is used to optimize the number of modes and the penalty factor to purify the BCG signal. By calibrating the characteristic peak points, the effective alignment of the BCG signal with the ECG signal was realized, laying the foundation for the subsequent calculation of HRV

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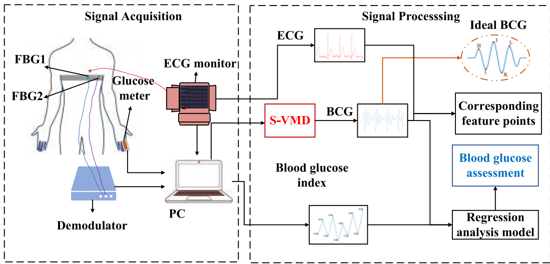

Fig. 1. Experimental system block diagram.

parameters. The correlation between the HRV of the BCG signal and the glycemic index was further analyzed, and the blood glucose level was inferred by monitoring the changes in HRV parameters. This method provides a new idea for noninvasive glucose monitoring.

II. Experimental Design

The block diagram of the experimental acquisition and processing of the BCG signal is shown in Fig. 1. It mainly contains two modules: signal acquisition and signal processing. The signal acquisition part includes the FBG sensor, noninvasive blood glucose detector, ECG, fiber optic demodulator, and computer, which is responsible for the acquisition, demodulation, and transmission of the signal. The signal processing part includes noise reduction and alignment analysis of the acquired ECG signal, BCG signal, and blood glucose values.

In the signal acquisition module, the FBG sensor is embedded in an elastic bandage and fixed to the human thoracic area. In order to distinguish the effects of temperature and strain on the center wavelength as much as possible, two FBGs were placed in this article for physiological signal detection. FBG1 collects heartbeat signals normally, while FBG2 is encapsulated with a rigid thin metal sheet for temperature compensation of body temperature. Avoid strain and temperature cross-sensitization issues. At the same time, an electrocardiographic lead was placed near the FBG1 position, and ECG signals from the same position were collected. The noninvasive glucose meter was clamped to the end of the index finger to record changes in blood glucose. When the sensing device receives the strain change, the amount of wavelength change captured by the thorax is transmitted into the demodulator for data demodulation and processing. For the signal processing module, first, we preprocessed the acquired signals (ECG, BCG, and blood glucose) to achieve initial noise reduction. Second, the BCG signal containing heartbeat information was purified by the S-VMD algorithm. Locate its feature points H, I, J, K, and L. Locate and analyze the above feature points one by one with the ECG signal feature points. Finally, the information from the alignment-free feature points is used to assess blood glucose levels through regression techniques to facilitate subsequent health monitoring.

III. Principle

A. VMD Algorithm

The VMD decomposes the original signal into a number of distinct intrinsic mode function (IMF) components. The center frequency and bandwidth of each component are determined by iteratively searching for the optimal solution of the variational model. The constrained variational model expression is given below

$$\left.\begin{aligned} &\min_{u_k,\omega_k} \sum_{k=1}^{K} \left|\partial t\left[\left(\delta(t)+\frac{j}{\pi t}\right)^{*} uk(t)\right] e^{-j\omega_k t}\right|; \\ &\text{s.t.} \sum_{k=1}^{K} u_k(t) = f(t). \end{aligned}\right\} \tag{1}$$

To solve the above equation, the quadratic penalization factor α and the Lagrange multiplier operator λ are introduced to solve the augmented generalized Lagrange function LL($\{uk\}$,$\{\omega k\}$,λ) of the variational problem with the following expression:

$$\begin{aligned} &L(\{u_k\},\{\omega_k\},\lambda) \\ &\quad= \alpha \sum_{k=1}^{k} \partial t\left[\left(\delta(t)+\frac{j}{\pi t}\right) * uk(t)\right] e^{-j\omega_k t} \\ &\qquad+ \left\| f(t) - \sum_{k=1}^{k} uk(t)\right\|_2^2 + \left[\lambda(t), f(t) - \sum_{k=1}^{k} uk(t)\right]. \end{aligned} \tag{2}$$

Using the alternating direction multiplier (ADMM) iterative algorithm, the computation is iterated repeatedly until the saddle point of the Lagrange function is found and the optimal solution is obtained.

B. S-VMD Algorithm

The VMD algorithm mainly relies on the number of decomposition layers K and the penalty factor α to determine the quality of signal decomposition. The value of K determines the number of modal components. If the value of K is too large, it will cause excessive decomposition of the signal, thus creating spurious modal components; if the value of K is too small, it will result in insufficient signal decomposition, thus mixing the modal component features with each other; When α is too large, it causes the feature signal to be discarded as noise; when α is too small, it causes the component to contain too much noise. The conventional center-frequency method [17] can only determine the value of K. The selection mostly relies on experience. If a complex optimization algorithm is chosen to determine the parameters, it increases the computational difficulty. Therefore, accurate selection of K and α is required to extract the effective signal from the original signal for analysis and reconstruction. To solve the parameter selection problem, this article proposes an adaptive selection algorithm to calculate the optimal parameter solution of VMD. In the decomposition process, only the center frequency of the last layer component is extracted, and the best K and α are automatically selected for VMD decomposition after meeting the set conditions, without the need for secondary operation. During the iteration process, following the changes in the number of decomposition layers and the bandwidth, the modal component is updated with the expression:

$$\mathrm{u}_{\mathrm{k}}^{(n+1)}(\omega) = \frac{f(\omega) - \sum_{k=1}^{k-1} \mathrm{u}_{\mathrm{k}}^{(n+1)}(\omega) + \sum_{k=k+1}^{K} \mathrm{u}_{\mathrm{k}}^{(n)}(\omega)}{1 + 2\alpha\left(\omega - \omega_{\mathrm{k}}^{(n)}\right)^2}. \tag{3}$$

The expression for the center frequency update is

$$\omega_{\mathrm{k}}^{n+1} = \frac{\int_0^{\infty} \omega \left|u_{\mathrm{k}}^{n+1}(\omega)\right|^2 d\omega}{\int_0^{\infty} \left|u_{\mathrm{k}}^{n+1}(\omega)\right|^2 d\omega}. \tag{4}$$

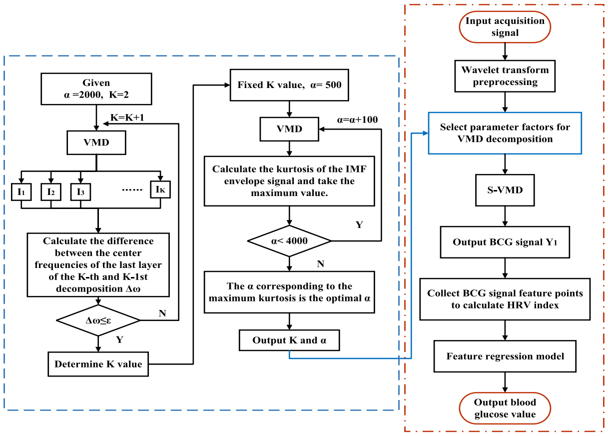

Fig. 2. S-VMD-based feature regression algorithm for neural networks.

The modal number K and the penalty factor α are selected for predecomposition in the following steps.

1) K is set to different values, the search range is [2], [12], and the value of the penalty factor α is fixed to 2000.
2) Calculate the frequency difference $\Delta\omega$ between the final components of neighboring K-values and take the absolute value, and also set the threshold value according to the frequency change. The expression is as follows:

$$\begin{cases} \Delta\omega = \left|\omega_{\mathrm{IMF}}^{\mathrm{K}+1} - \omega_{\mathrm{IMF}}^{\mathrm{K}}\right| \\ = \left| \dfrac{\int_0^\infty \omega_{\mathrm{k}+1} \left|u_{\mathrm{IMF}}^{\mathrm{K}+1}(\omega)\right|^2 d\omega}{\int_0^\infty \left|u_{\mathrm{IMF}}^{\mathrm{K}+1}(\omega)\right|^2 d\omega} - \dfrac{\int_0^\infty \omega_{\mathrm{k}} \left|u_{\mathrm{IMF}}^{\mathrm{K}}(\omega)\right|^2 d\omega}{\int_0^\infty \left|u_{\mathrm{IMF}}^{\mathrm{K}}(\omega)\right|^2 d\omega} \right| \le \varepsilon \\ \varepsilon = \dfrac{\sum_{i=1}^{n} \Delta\omega_{\mathrm{K}}}{n} \\ \text{s.t.} \begin{cases} \mathrm{K} \in [2, 12] \\ i = 1, 2, \ldots, \mathrm{K} \\ \alpha = 2000 \end{cases} \end{cases} \tag{5}$$

3) ε is the set threshold. When $\Delta\omega \le \varepsilon$, K is selected as the optimal solution. For the setting of the threshold, the spectral kurtosis value of each mode is iteratively calculated, and the decomposition is terminated when the spectral kurtosis increment of the newly added mode is less than 5% according to the kurtosis maximization criterion [18]. This threshold is derived based on the Shannon–Nyquist sampling theorem to ensure that more than 95% of the signal energy is retained.
4) Fixed K value, initialize α value to 500, step size 100, search range [500, 4000].
5) Calculate the peak envelope signal index $\mathrm{EK}_{K_i}^{\alpha}$, for each IMF component with a fixed α until the end of the loop [19], [20]. The kurtosis indicator is calculated using the following formula:

$$\begin{cases} \mathrm{EK}_{\mathrm{K}_i}^{\alpha}(i) = \dfrac{E\left(E_{\mathrm{K}_i}^{\alpha} - \mu\left(E_{\mathrm{K}_i}^{\alpha}\right)\right)^4}{\left(\sigma\left(E_{\mathrm{K}_i}^{\alpha}\right)\right)^4} \\ \text{s.t.} \begin{cases} \alpha \in [500, 4000] \\ i = 1, 2, \ldots, \alpha \\ \mathrm{K}. \end{cases} \end{cases} \tag{6}$$

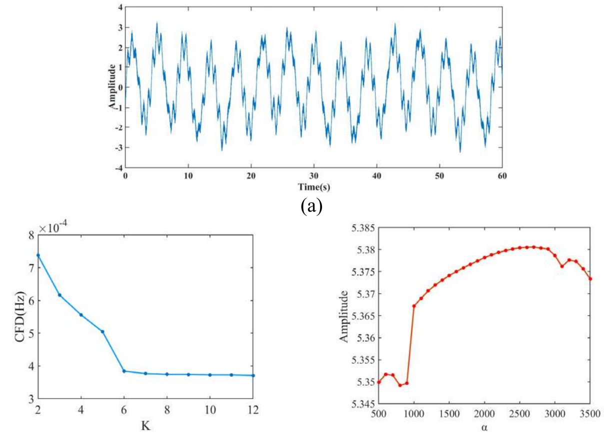

Fig. 3. Simulated signals and processing effects. (a) Simulation signal. (b) Difference in the center-frequency value of the last IMFK. (c) Envelope signal kurtosis values of each IMF component for different α.

When α is fixed, the peak size of each IMF component envelope signal is compared to obtain V. When fixed, the peak signal size of each IMF component envelope is compared with obtain $E^{\max} K_K^{\alpha}$ called the local maximum. Compare each local maximum and consider the α corresponding to max $E^{\max} K_K^{\alpha}$ as the best α value. The flowchart of the algorithm is shown in Fig. 2.

IV. Simulation and Experimentation

A. Simulation Verification

To verify the feasibility of the algorithm proposed in this article, the heartbeat signal is simulated in MATLAB by coupling sinusoidal signals with reference to the literature [21], [22]. Since external interference is unavoidable in practical measurements. To increase the level of signal realism, we add breathing simulation, baseline drift, and 20 dB [21]. Gaussian white noise to the simulated signal. Baseline drift is added to simulate external temperature, and Gaussian white noise to simulate high-frequency noise and friction vibration. The simulated signal expression is

$$\begin{cases} y_1 = A_1 \sin(2\pi\omega_1 t) \\ y_2 = \sum_{n=1}^{3} A_i \sin(2\pi\omega_i t + \varphi_i) \\ y_3 = A_3 \sin(2\pi f_c t) \\ y = \mathrm{awgn}((y_1 + y_2 + y_3), 20). \end{cases} \tag{7}$$

In the expression, y_1 is the respiratory signal and y_2 is the heartbeat signal. A_i is the amplitude and ω_i is the frequency, which are taken from the experimental results of [21]. y_3 is the baseline drift. f_c is the random frequency of [200, 1000]. $y$is the synthesized signal with added noise. With a sampling frequency of 1KHz and a time of 60 s. The simulation results are shown in Fig. 3(a).

The simulated signal is decomposed using the VMD algorithm to obtain the IMF component. Fig. 3(b) shows the difference in the center frequencies of the endmost components for different K-values. The change of $\Delta\omega$ starts to level off, and after several sets of experiments on the simulated signals, it is found that when $\Delta\omega \le 0.5\mathrm{e}^{-04}$, the change in $\Delta\omega$ starts to level off, and the value of K can be determined. After fixing

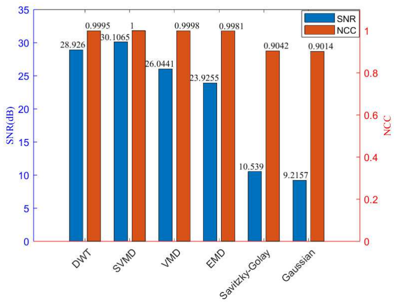

Fig. 4. Evaluation indicators and calculations.

K, the S-VMD decomposition of the signal is performed by changing the value of α. The stratified kurtosis metrics for different α values after removal of the respiratory signal and baseline drift components are shown in Fig. 3(c).

In order to verify the quality of the noise reduction signal, signal-to-noise ratio (SNR) and normalized correlation coefficient (NCC) are selected as the evaluation indices in this article

$$\mathrm{SNR}\,(\mathrm{dB}) = 10 * \lg\left(\frac{P_{\text{signal}}}{P_{\text{noise}}}\right) \tag{8}$$

$$\mathrm{NCC} = \frac{\sum_{n=1}^{N} As\,(n)\,Ad\,(n)}{\sqrt{\left(\sum_{n=1}^{N} As^2\,(n)\right)\left(\sum_{n=1}^{N} Ad^2\,(n)\right)}}. \tag{9}$$

SNR is the ratio of signal-to-noise power. P_{signal} is the denoising signal power, and P_{noise} is the noise power. Larger SNR values indicate better noise reduction. NCC reflects the overall similarity of the signal waveform before and after denoising. As refers to the pure signal, and Ad refers to the noise reduction processed signal. The closer the NCC value is to 1, the closer the two waveforms are. Other filtering noise reduction methods are introduced, such as Gaussian, Savitzky–Golay filtering, EMD, DWT, and conventional VMD algorithms. The SNR and NCC of the signal after noise reduction of the above five methods and the proposed algorithm in this article are calculated, respectively. The results are displayed as shown in Fig. 4.

It can be seen that the signal after noise reduction based on the S-VMD algorithm has an SNR value of 30.1065 with the pure signal, and the NCC is 1. It shows that the algorithm used in this article has the best denoising effect for BCG signals with the highest degree of similarity to the original signal.

B. Experimental Acquisition

The signal acquisition device consists of FBG1 (heartbeat), FBG2 (temperature), fiber optic demodulator, ECG machine, noninvasive glucose meter, and PC. The experimental volunteers were 30 youths aged 22–27 years, 15 females, 15 males. Due to the complexity of diabetic health conditions and the fact that the initial concept of this experiment is geared toward the control and detection of blood glucose in everyday life for the average person. Therefore, the volunteers were recruited in good health, with no history of triple hypertension, insulin medication, or cardiovascular disease. Smoking, drinking alcohol, and exposure to caffeine-containing foods were avoided for 12 h before the start of the experiment. The data on the physical characteristics of the volunteers are shown in Table I.

TABLE I
VOLUNTEER POPULATION STATISTICS

Variable	Test
Number	30
Age(years)	23±3
Sex(%female)	15(50)
BMI (kg/m^2)	21.8±1.8
Heart rate(bpm)	78±9
Duration of diabetes (years)	N/A
Fasting blood glucose(mmol/L)	4.35±0.67
Postprandial blood glucose(mmol/L)	6.16±0.8
Current smoker(%)	3(10)

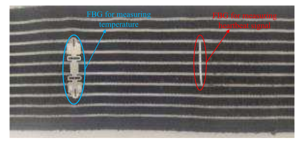

Fig. 5. Sensor embedding position diagram.

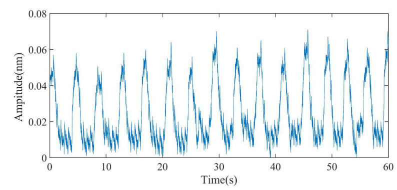

Fig. 6. Experimental signal collection diagram.

The changes in fasting blood glucose levels after 8 h of fasting and heart rate variability ballistocardiography (HRVBCG) parameters after 1 h of feeding were tested in Chi volunteers.

According to the physiological structure of the human body, the collection location of the heartbeat signal in this experiment is 4 cm to the left of the sternal raphe. The FBG fiber is embedded in the elastic chest strap by hand-stitching, and no other fillers are added to ensure its monitoring accuracy. The sensor embedding schematic is shown in Fig. 5.

The experiments were carried out in a constant temperature environment. The FBGS were fixed at the chests of the volunteers, the ECG machine connected the leads to the corresponding parts of the body in turn, and the noninvasive glucose meter was clamped to the index finger of the body. FBG1 is a reference grating that only detects temperature changes and is not affected by pressure. As a sensor grating, FBG2 is affected by both pressure and temperature. Temperature compensation is achieved by subtracting the wavelengths of the two reflection centers to eliminate the effect of temperature changes on the pressure measurement [23].

Before the start of the experiment, we performed a standardized calibration of all sensors to test the response of the device in the target frequency band (0.5–10 Hz) by inputting an analog signal of known frequency. Calibration was performed every 12 h to ensure that the sensors were always performing optimally. The experiments were timed using a stopwatch, and all sensor data recording times were synchronized with the experimental operations to ensure that the data had an error-free temporal relationship to the experimental operations. After the volunteers calmed down, blood glucose was tested on the same finger using noninvasive and invasive glucose meters, respectively, and the values were recorded. To avoid adverse effects caused by the long-term connection of ECG lead balls

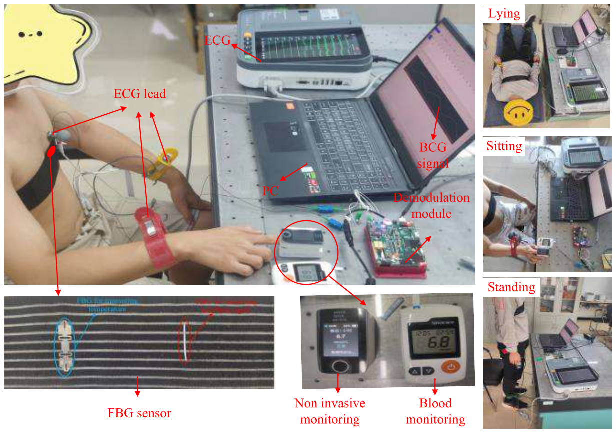

Fig. 7. Overall experimental monitoring.

to multiple human bodies, the experiment duration is set to 1 minute, and the signal acquisition status is as shown in Fig. 6. The laboratory environment, including temperature (25 °C ± 3 °C), humidity, and light, was strictly controlled during the experiment to ensure that the interference of the external environment on the measurement results was minimized. Volunteers were monitored experimentally in three states: sitting, standing, and lying down. The chest strap with embedded FBG was connected to the fiber optic demodulator, and the sampled signals were recorded and output at a sampling frequency of 1 kHz. The actual monitoring is shown in Fig. 7.

C. Feature Extraction

The Symlet8 wavelet basis [24] is chosen for the raw signals captured to perform a 10-layer decomposition to achieve detrending. Taking the data taken by volunteer no. 3 as an example, the VMD decomposition parameters K = 5 and α = 2800 can be calculated. Subsequently, the preprocessed signal is purified by secondary noise reduction. The useful signals are selected based on the correlation coefficients and spectra of the decomposed components, and the rest are considered as noise and rounded off. Finally, the extracted and separated BCG signals and ECG signals are aligned by the differential thresholding method for feature point localization. The comparison signals are normalized for ease of observation. The comparison results are shown in Fig. 8.

Due to the complex mechanical properties of the heart's contraction and diastole. The acquired weak vibration signals of the human body caused by the contraction and expansion of the heart are different from the ECG signal-acquired myocardial electrical signal waveforms, but both show regular changes. The HRV parameters of both were also calculated. HRV parameter estimation mainly consists of time domain analysis and frequency domain analysis. In the time domain, the RR interval (normal to normal heartbeat), the standard deviation of the RR interval SDNN, the mean square deviation of consecutive RR intervals root mean square of successive difference (RMSSD), and the percentage of adjacent RR intervals differing by more than 50 ms PNN50 are included. In the frequency domain, it contains low-frequency LF (0.04–0.15 Hz), high-frequency HF (0.15–0.4 Hz), and the ratio of low-to-high-frequency power, LF/HF. However, LF and HF are usually discussed in the context of continuous 5-min data. In this article, because of the effect of prolonged

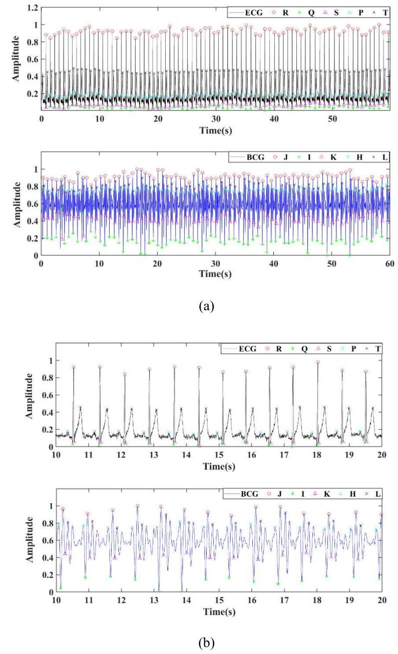

Fig. 8. Comparison of BCG signal and ECG signal. (a) 1-min signal. (b) 10–20-s signal.

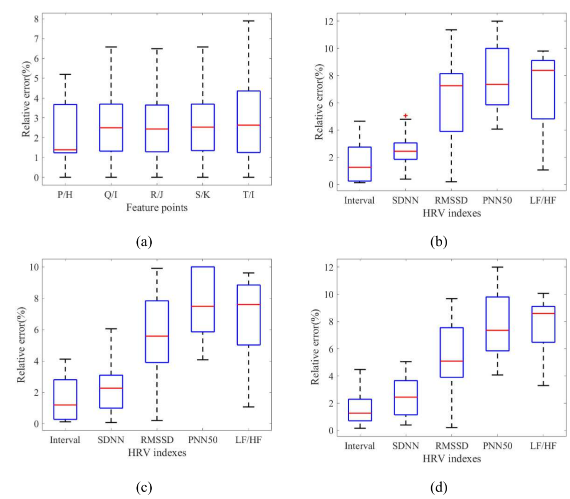

Fig. 9. Box plots of relative errors between BCG and ECG feature data. (a) Box plots of characteristic point errors. (b) Box plots of H and P point HRV index errors. (c) Box plots of J and R point HRV index errors. (d) Box plots of L and T point HRV index errors.

contact of the cardiac lead ball with the skin, the sampling time of each set of data was 1 min. Therefore, the analytical value of LF, HF, and LF/HF is reduced. The relative error analysis of the BCG signal and ECG signal feature points is shown in Fig. 9.

Fig. 9 shows the BCG signal and ECG signal characteristic data error box plots.

In Fig. 9(a), the median relative errors of the characterized points are P–H: 1.39%, Q–I: 2.5%, R–J: 2.44%, S–K: 2.53%, and T–I: 2.63%. The average relative errors were less than 3%. It shows that the localization and identification of the characteristic points of BCG and ECG signals are basically in agreement. To further verify the feature correlation between BCG and ECG signals, the HRV parameters of BCG signal I,

TABLE II
HRV Parameter Calculation Results

Parameter	BCG (fasting)	BCG (after meals)
HR(bmp)	90±4	76±5
Interval(ms)	633.21±33.21	786.92±20.55
SDNN (ms)	42.15±10.33	36.5±7.8
RMSSD (ms)	36.12±12.31	25.67±10.31
PNN50	8.7±4.5	6.7±5.1
LF/HF	1.6±1.5	1.2±1.1
Blood glucose	4.62±0.31	6.56±0.54

J, and K feature points and Q, R, and S feature points of the corresponding ECG signals were calculated, respectively.

In Fig. 9(b), the median relative errors of the HRV parameters of the H-wave of the BCG signal and the P-wave of the ECG signal are as follows. Interval: 1.2647%, SDNN: 2.4429%, RMSSD: 7.4715%, PNN50: 7.3559%, and LF/HF: 8.3854%. The average relative error is less than 10%.

In Fig. 9(c), the median relative errors of the HRV parameters of the main wave J-wave of the BCG signal and the main wave R-wave of the ECG signal are as follows. Interval: 1.1935%, SDNN: 2.2617%, RMSSD: 7.2589%, PNN50: 7.4837%, and LF/HF: 7.5956%. The average relative error is less than 8%.

In Fig. 9(d), the median relative errors of the HRV parameters of the L-wave of the BCG signal and the T-wave of the ECG signal. Interval: 1.3819%, SDNN: 2.4517%, RMSSD: 5.0887%, PNN50: 6.9859%, and LF/HF: 8.5932%. The average relative error is less than 10%.

Based on the above data analysis, a strong correlation is observed between ECG and BCG characteristic points in terms of interval measurements. The calculated HRV parameters demonstrate an average relative error within 10%. According to previous studies [9], [10] that established the correlation between ECG-derived HRV and blood glucose variations, the HRV parameters extracted from BCG signals can also be effectively applied for noninvasive blood glucose monitoring.

D. Analysis of HRV Parameters and Glycemic Changes

To observe the distribution of BCG signal HRV parameters and blood glucose changes. HRV parameters were calculated from BCG signals, and blood glucose values were collected by the Sanno GA-3 minimally invasive blood glucose meter. The blood glucose meter is an invasive test with a measurement range of 1.1–33.3 mmol/L and a response time of 10 s. The data are shown in Table II.

Monitoring data revealed that compared with fasting-state measurements, postprandial blood glucose levels exhibited significant fluctuations accompanied by corresponding changes in HRV parameters. Specifically, elevated blood glucose levels were associated with increased heart rate and decreased values of SDNN, RMSSD, and PNN50 compared with fasting conditions. Using minimally invasive blood glucose meter readings as reference values, we established a multiple linear regression model to characterize the relationship between blood glucose levels and HRV parameters. The model formulation is as follows:

$$BL = \beta_0 + \beta_1 \text{Interval} + \beta_2 \text{SDNN} + \beta_3 \text{RMSSD} + \beta_4 \text{PNN50} + \beta_5 (\text{LF/HF}) + \varepsilon. \quad (10)$$

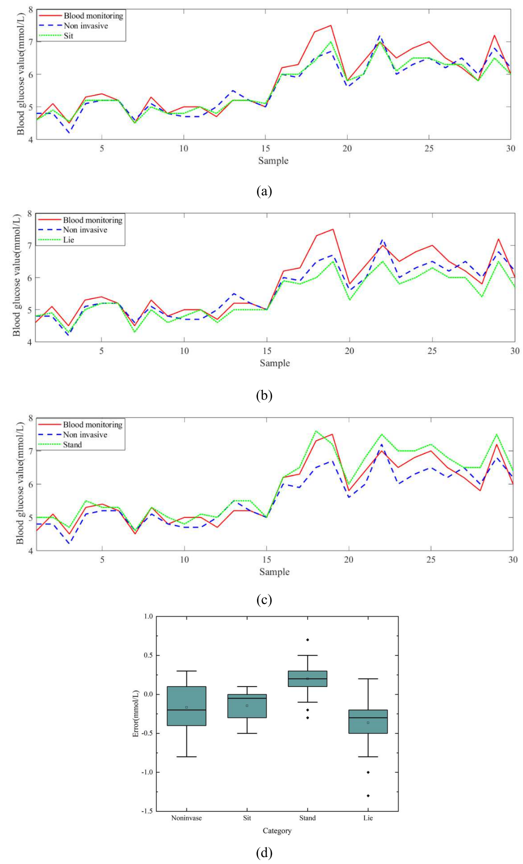

Fig. 10. Analysis of the mapping relationship between blood glucose values and HRV parameters. (a) Blood glucose test value (Sit). (b) Blood glucose test value (Lie). (c) Blood glucose test value (Stand). (d) Error box plot.

In the equation, BL represents the dependent variable (blood glucose level), while Interval, SDNN, RMSSD, PNN50, and LF/HF serve as independent variables (HRV parameters). β_0 denotes the intercept, β_1 through β_5 are regression coefficients, and ε represents the error term.

In this article, the minimally invasive glucose meter monitoring value is the real value of blood glucose, and the simultaneous collection using the Bobang Ark noninvasive glucose meter is used as the comparison value. The fitting changes are shown in Fig. 10.

Fig. 10(a)–(c) shows the contrasting values of minimally invasive, noninvasive, and modeled glucose monitoring in three states of sitting, standing, and lying down, respectively, and Fig. 10(d) shows the box plots of the absolute error of the glucose measurements and the true values. By organizing and analyzing the experimental data, it was found that the correlation between the true value and the noninvasive monitoring value was 0.85, and the mean square error (mse) was 0.13. In the sitting state, the correlation between the true value and the model-predicted value was 0.96, and the mse was 0.08. In the standing state, the correlation was 0.92, and the mse was 0.10. In the lying state, the correlation was 0.89, and the mse was 0.11. The data was analyzed in the box plot of Fig. 10(d). The data indicate that there is a strong correlation between changes in blood glucose values and changes in HRV parameters, whether sitting, standing, or lying down. The overall correlation is greater than 0.85, and the mse is less

TABLE III
RESULTS OF HRVBCG COMPARED WITH OTHER NONINVASIVE TECHNIQUES

Parameters	HRVBCG	ECG	PPG	IR
Accuracy MSE	≥85% $<$0.12	≥90% ≤0.1	≥82% $<$0.2	≥90% ≤0.15
Hardware Complexity	Low Integrated mattress/seat	Medium (electrode patch)	Low (wrist-worn)	High (specialized equipment)
Temperature	-10 ~ 50°C	15 ~ 35°C	20 ~ 30°C	20 ~ 30°C
Practicality	High (long-term monitoring)	Low (electrodes to be applied)	High (wearable)	Low (lab environment)
Cost	Low	High	Low	High

than 0.12. In the standing position, gravity causes more blood to flow to the lower limbs, whereas when lying down, the blood flow slows down, which affects the monitoring effect. Therefore, the correlation between the measured value and the true value was the highest, and the mse was the lowest in the sedentary state. It shows that it is feasible to study HRVBCG for subsequent noninvasive monitoring of blood glucose. The results of the comparison of HRVBCG with other noninvasive techniques (e.g., PPG, ECG, and IR spectroscopy) in terms of key metrics such as accuracy, sensitivity, hardware complexity, and utility are shown in Table III.

V. DISCUSSION

The noninvasive blood glucose monitoring method based on fiber optic sensing HRVBCG proposed in this study requires no complex optics and precision calibration, has lower hardware cost, and is easy to integrate into wearable devices compared with traditional methods such as metabolic heat [7] and infrared spectroscopy [8]. Compared with methods based on PPG [9] and ECG [10], [11], BCG monitoring is environmentally adaptable, has less influence on temperature and humidity, is less binding, and can synchronize the collection of physiological parameters such as heart rate and respiration.

However, there are still some limitations to the methodology. First, the sample of research subjects should be more diverse, and it is recommended that subjects of different age groups and blood glucose levels be included through multicenter clinical studies to ensure a balanced sample. Second, the experimental setting is relatively homogenous and does not adequately consider external factors such as environmental changes, mood swings, and physical activity. There are challenges in integrating the technology into wearable devices, where sensor stability and real-time data processing capabilities are critical, and ensuring that the devices work efficiently over long periods is an issue. Finally, the robustness of the algorithms still needs to be improved to cope with complex and noisy signals.

Future studies should validate the applicability of the method in different populations by expanding the sample size, especially by including diabetic patients and volunteers of different age levels. Meanwhile, the stability and accuracy of the system need to be tested in more complex real-world environments. By optimizing the hardware design and enhancing the data fusion processing capability. To realize a "miniaturized" device that integrates monitoring, demodulation, and output. In addition, the combination of other physiological parameters (e.g., blood oxygen, blood pressure, etc.) may provide users with more comprehensive health monitoring and improve the accuracy and convenience of health management. Overall, despite the current technical bottlenecks, this method has great potential in wearable health monitoring systems and is expected to realize real-time monitoring of blood glucose around the clock in the future, providing a more accurate and convenient solution for individual health management.

VI. CONCLUSION

In this article, a fiber optic sensing-based HRVBCG noninvasive glucose monitoring study is presented. This study focuses on two main aspects: one is the feature extraction of BCG signals based on fiber optic sensing. To address this problem, a VMD parameter selection algorithm based on the center frequency and peak index is proposed to highly preserve the feature data of BCG signals. Compared with the ECG signal, the average relative error of each feature point was less than 3%, and the average relative error of each parameter was less than 8% for both HRVBCG and HRVECG. It showed that the BCG signal and ECG signal localization coincided, indicating that the BCG signal and ECG signal have the same reference significance in physiological monitoring. On the other hand, the change in blood glucose values was analyzed in correlation with the amount of change in HRV. For this article, healthy volunteers with no history of related diseases were recruited to monitor the amount of change in blood glucose values and changes in HRV parameters after fasting and eating. It was found that when the blood glucose values increased, the parameters such as SDNN and RMSSD started to decrease. Also, the volunteers' heart rates were elevated compared with when they were fasting. The overall correlations between the parameter-fit regression models and the amount of change in blood glucose values were all greater than 0.85, and the MSEs were all less than 0.12. It can provide a more convenient and economical way in the field of noninvasive blood glucose monitoring.

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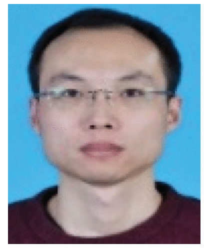

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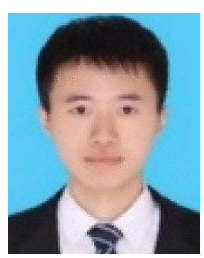

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